Dashboard 1: Loan Risk Analysis

Introduction:

A comprehensive understanding of loan performance and associated risks is achieved through an in-depth analysis of key financial and demographic indicators such as loan statuses, default patterns, borrower profiles, and creditworthiness metrics. This helps uncover trends, assess portfolio health, and identify potential areas of financial risk. The analysis, developed using the preprocessed Bondora dataset, brings these insights together by outlining the main objectives, key performance indicators (KPIs), interactive filters, and slicers. It also includes the essential formulas used in the calculations, enabling a clear and data-driven evaluation of loan behavior and borrower reliability.

DAX Queries:

1. **Total Loans Issued:**

```
Total_Loans_Issued = DISTINCTCOUNT(bondora_data[loanid])
```

2. **Average Loan Amount**:

```
Average_Loan_Amount = AVERAGE (bondora_data[amount])
```

3. **Default Rate (%)**:

```
Default Rate (%) =
VAR TotalLoans = COUNTROWS (bondora_data)
VAR DefaultedLoans = COUNTROWS (FILTER (bondora_data,
bondora_data[status] = "Late"))
RETURN
DIVIDE (DefaultedLoans, TotalLoans, 0)
```

4. **Average Probability of Default (PD)**:

```
Average PD = AVERAGE (bondora_data[probabilityofdefault])
```

5. **Expected Loss (EL)**:

```
Total_Expected_Loss = SUM (bondora_data[expectedloss])
```

6. **Recovery Rate (%)**:

```
Recovery Rate (%) =
DIVIDE(
SUM(bondora_data[principalpaymentsmade]),
SUM(bondora_data[principaloverduebyschedule]) +
SUM(bondora_data[principalbalance]),
0
)
```

Visualizations:

Visualization	Visualization Type
Total Loans Issued	KPI Card
Average Loan Amount	KPI Card
Default Rate (%)	KPI Card
Average PD	KPI Card
Expected Loss (EL)	KPI Card
Recovery Rate (%)	KPI Card
Loans by Status	Donut Chart • showing proportions of each **loan status**
Loans by Credit Rating	**Bar Chart** • X-axis: sum(amount) • Y-axis: rating
Expected Loss vs. Expected Return	**Clustered Bar Chart** • X-axis:Sum(expectedloss), Sum(expectedreturn) • Y-axis: useofloan
Geographic Distribution of Loans	**Filled Map:** • **Location:** Country • **Legend:** Country
Loan Purpose Breakdown	**Tree Map:** • **Category:** Useofloan • **Values:** Sum(amount)

Formulas/Calculations:

Calculation	Formula (DAX)
Defaulted Loans Count	`Defaulted Loans Count = COUNTROWS(FILTER(bondora_data,bondora_data[status] = "Defaulted"))`
Active Loans Count	`Active Loans Count = COUNTROWS (FILTER (bondora_data, bondora_data[status] = "Current"))`
Total Loan Amount	`Total Loan Amount = SUM(bondora_data[amount])`
Debt-to-Income Ratio (DTI)	`DTI = AVERAGE(bondora_data[debttoincome])`
Loans in Late Category (%)	`Loans in Late Category (%) = DIVIDE (COUNTROWS (FILTER (bondora_data, bondora_data[worselatecategory] <> "None")), COUNTROWS(bondora_data))`

Filters:

- **Loan Status**
- **Country**
- **Loan Duration (**(e.g., 6, 12, 36 months)**)**
- **Credit Score (Rating)**
- **Employment Status**
- **Use of Loan**

Slicers:

- **Loan Application Date Range (ListedOnUTC):** Select loans issued within a specific time period.
- **Age Range:** Slicer to filter loans by the borrower's age group (e.g., 18–25, 26–35, etc.).
- **Loan Amount Range (Amount):** Filter based on the amount of the loan issued.

Expected User Actions:

- **Interactively explore the data:** Utilize slicers (e.g., date range, loan amount range, credit rating) to dynamically filter and examine trends across different segments.

- **Identify risk patterns:** Analyze the distribution of loans across categories such as age group, credit rating, and loan purpose to uncover areas with elevated default risk and potential vulnerabilities in the portfolio.
- **Evaluate recovery effectiveness:** Compare recovery rates against default rates to assess the efficiency of debt recovery processes, highlighting opportunities to improve collections strategies and reduce financial losses.

Conclusion:

Analysis of **loan performance** highlights key insights into **borrower behavior** and associated **risks**, including **total loans issued**, **average loan amounts**, **default rates**, **expected losses**, and **recovery efficiency**. Interactive exploration by **age**, **credit rating**, **loan purpose**, and **geography** reveals **high-risk segments** and trends in **defaults** and **recoveries**. These insights enable informed decisions for improving **debt recovery strategies**, mitigating **risk exposure**, and optimizing **portfolio performance** across diverse **borrower categories**.